

Space Mission Design & Operations

3-Week Intensive Programme — Syllabus

SpaceCDF Course

Summer 2026

Audience and pre-requisites

This programme suits **industry professionals, graduate students, and upper-year undergraduates** with an engineering background and the appetite to make decisions quickly and in public. We assume basic mechanics, electromagnetism, and programming literacy — but **no prior spacecraft experience**. The cohort runs **20 to 30 participants** in **interdisciplinary teams of 4 to 5**.

The course is delivered Monday to Friday, 09:00 – 16:00, in person at the University of Ottawa. The tool spine is the **SpaceCDF AI-assisted Concurrent Design Facility**.

EXPECTED READING

Expected reading NASA CubeSat 101 (≈ 90 min) — https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf. NASA Systems Engineering Handbook §2 (≈ 60 min) — <https://www.nasa.gov/reference/systems-engineering-handbook/>.

How the three weeks fit together

Project Lifecycle Phases & Review Gates — NASA vs ECSS

NASA gates

MCR SRR SDR PDR CDR SIR ORR FRR DR

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MDR PRR SRR PDR CDR QR AR FRR ELR

ECSS gates

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Figure 1.1 — The lifecycle the cohort lives. Week 1 grounds you in Pre-Phase A; Week 2 walks the Phase A/B portion of the V; Week 3 puts you in operator seats during commissioning.

The course follows the **System-V model** — decomposition on the left, integration and verification on the right — and aligns each day to a recognisable lifecycle phase.

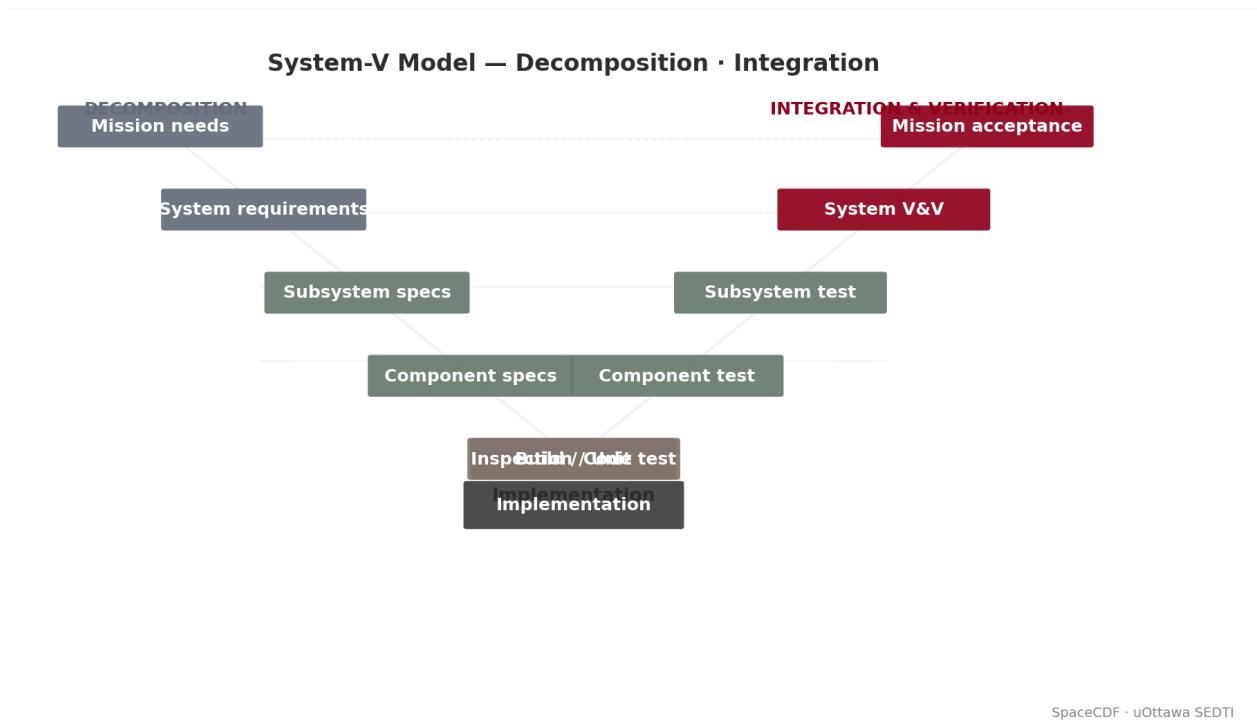
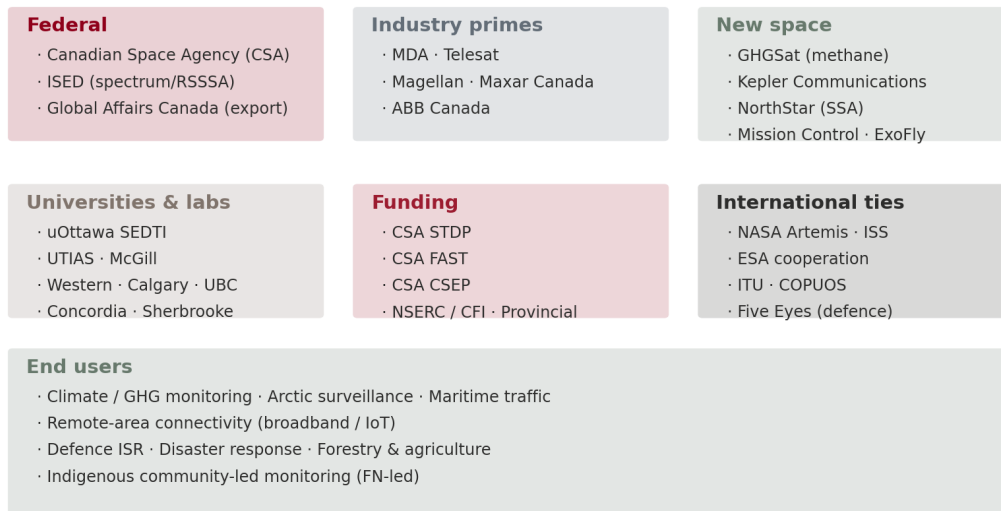


Figure 1.2 — System-V with course touchpoints. Mission-needs work happens Day 1; subsystem specs converge Days 2–3; integration and V&V are Day 4; the gate review is Day 5.

Week 1 — Canadian Space Landscape & Regulatory Environment

This week grounds the cohort in the system the missions live in: the **Canadian space ecosystem**, the **regulatory landscape**, and the systems-engineering frameworks that professionals actually use. Lectures are deliberately short — the aim is to set context for the tool-led work that begins in Week 2.

The Canadian Space Sector — illustrative map of actors



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Figure 1.3 — A map of the Canadian space sector. Where students see themselves in this picture is the central question of Day 1.

DAILY SCHEDULE

Day	Topic	Activities
Mon	The Canadian Space Ecosystem	Canadian Space Agency mandate & programmes. Industry map: MDA, Telesat, GHGSat, Kepler, NorthStar, university labs. CSA funding (STDP, FAST, CSEP). Canada's role in Artemis, Lunar Gateway, Radarsat. Guest speaker (CSA or industry).
Tue	Regulatory Framework	ISED spectrum licensing under CPC-2-6-02. RSSSA (Remote Sensing Space Systems Act). Canadian Controlled Goods Programme and export control. ITU filing through ISED. Practical: complete an ISED spectrum-licence application for a sample mission.
Wed	International Standards & Compliance	ECSS standard framework and tailoring. NASA SEH 17 processes overview (see Figure 2). Cal Poly CDS Rev 14. Space-debris mitigation (IADC 25 yr, FCC 5 yr). UN Registration Convention. Practical: complete a tailoring matrix.
Thu	Mission Needs & Opportunities	Identifying mission needs in the Canadian context (Arctic monitoring, maritime surveillance, agriculture, resource management, connectivity). Stakeholder analysis. Indigenous community engagement and UNDRIP. Problem-statement workshop.
Fri	Space vs Non-Space Trade & Concept	Trade-study methodology (criteria, weightings, scoring). Existing services (Radarsat, Sentinel, Planet) vs new missions. Constellation economics and business cases. Teams form, select a mission need, and run an initial trade analysis in SpaceCDF.

CANADIAN REGULATORY CHECKLIST (TUESDAY DELIVERABLE)

Regulator	Filing	Trigger	Approximate timeline
ISED (Spectrum Management)	Radio licence — CPC-2-6-02	Any RF emission to/from satellite	6 – 12 months
ISED (RSSSA)	RSSSA licence	Earth-observation payload above resolution threshold	12 – 18 months
ITU (via ISED)	API → coordination → notification	New frequency assignment	2 – 5 years
Global Affairs Canada	Export permit (ECL / CGP)	Cross-border hardware, data, technology transfer	4 – 12 weeks
UNOOSA (via Global Affairs)	Registration of Space Object	Launch into orbit	At launch
ISED (Space-Object)	Debris-mitigation plan	Launch — IADC 25 yr / FCC 5 yr	At launch

STANDARD REFERENCE

Standard reference ECSS-E-ST-10C Rev. 1 — System engineering general requirements — <https://ecss.nl/standards/active-standards/ecss-e-st-10c-rev-1-system-engineering-general-requirements/>. NASA SEH (SP-2016-6105 Rev 2) — <https://www.nasa.gov/reference/systems-engineering-handbook/>.

MISSION ARCHETYPES USED IN WEEK 1 TRADE STUDIES

Archetype	Orbit	Payload	Comms	Typical use
Ocean-colour EO	450 – 700 km SSO, ≈ 98°	Multispectral imager	S-band	GHG, climate, fisheries
SAR EO	500 – 700 km SSO	C/X/Ka SAR	X-band	Maritime, ice, defence
AIS	400 – 600 km LEO	RF receiver	UHF/VHF	Maritime traffic
IoT / data relay	500 – 1100 km LEO	Store-forward radio	UHF/Ka	Connectivity, IoT
Climate-GHG	500 km SSO	Spectrometer (point)	S-band	Methane, CO ₂
LEO-PNT	500 – 1200 km MEO/LEO	GNSS supplement	L-band	Backup PNT
Constellation comms	500 – 1200 km LEO	Phased-array	Ka-band	Broadband/IoT

WEEKEND PRE-WORK (OVERNIGHT READING FLAG)

EXPECTED READING

Expected reading ECSS-M-ST-10C Rev. 1 §5 (project-management lifecycle), ≈ 30 min — <https://ecss.nl/>. NASA CubeSat 101 §3 (mission types).

Expected reading before Wednesday. Cal Poly CubeSat Design Specification Rev 14, §1 – §3 (≈ 45 min) — <https://www.cubesat.org/cds-announcement>.

Expected reading before Friday. SpaceCDF user guide §1 – §3 (≈ 30 min). Bring a one-page mission idea on Friday morning.

Week 2 — Concurrent Design Facility — Designing a Mission

Week 2 is the **CDF intensive**. Each interdisciplinary team converges on a complete preliminary mission design across five working days, supported by SpaceCDF’s embedded systems-engineering process, risk register, and trade-study modules. The PDR sits at end of week.

Mission Operations Timeline (representative LEO mission)

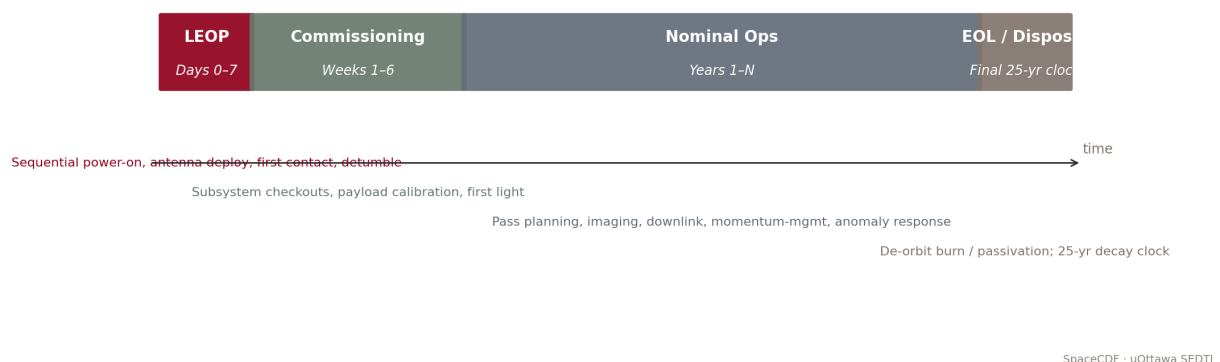


Figure 1.4 — A representative LEO mission timeline. Week 2 builds the design that will fly through this timeline.

DAILY SCHEDULE

Day	Topic	Activities
Mon	System-V & Requirements	System-V model deep dive. SMART requirements from objectives. Functional decomposition. Teams enter mission need, run orbit/class advisors, generate the requirements baseline in SpaceCDF.
Tue	Subsystem Design: Power, AOCS, Thermal	Orbit selection trade. Solar-array and battery sizing. Duty cycling. Attitude-control selection. Thermal environment analysis. Teams run the design loop and review parametric budgets.
Wed	Subsystem Design: Comms, Structure, Propulsion	Link-budget calculation. Frequency-band selection and licensing implications via the SpaceCDF regulatory check. CubeSat structure and deployer selection. Propulsion trade. Equipment selection in the SpaceCDF library.
Thu	Integration, Verification, Cost	Interface-matrix review. V&V matrix assignment. Environmental test planning. Cost estimation (parametric + COTS). Bill of materials. Trade studies on key decisions.
Fri	Design Review & Optimisation	Run optimiser. Sensitivity analysis. Gate review (PDR criteria). Resolve conflicts. Generate ECSS documents (MRD, TS, VP). Regulatory filings (ITU, RSSSA if applicable). Each team presents its preliminary design.

A FIRST LOOK AT THE BUDGETS YOU WILL CLOSE

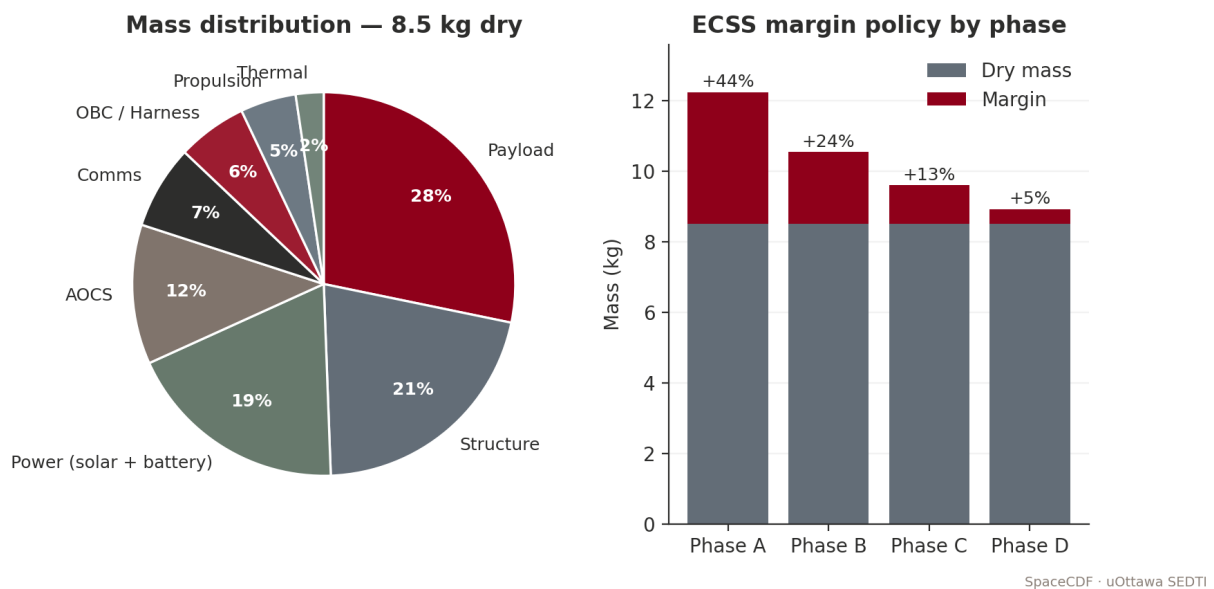
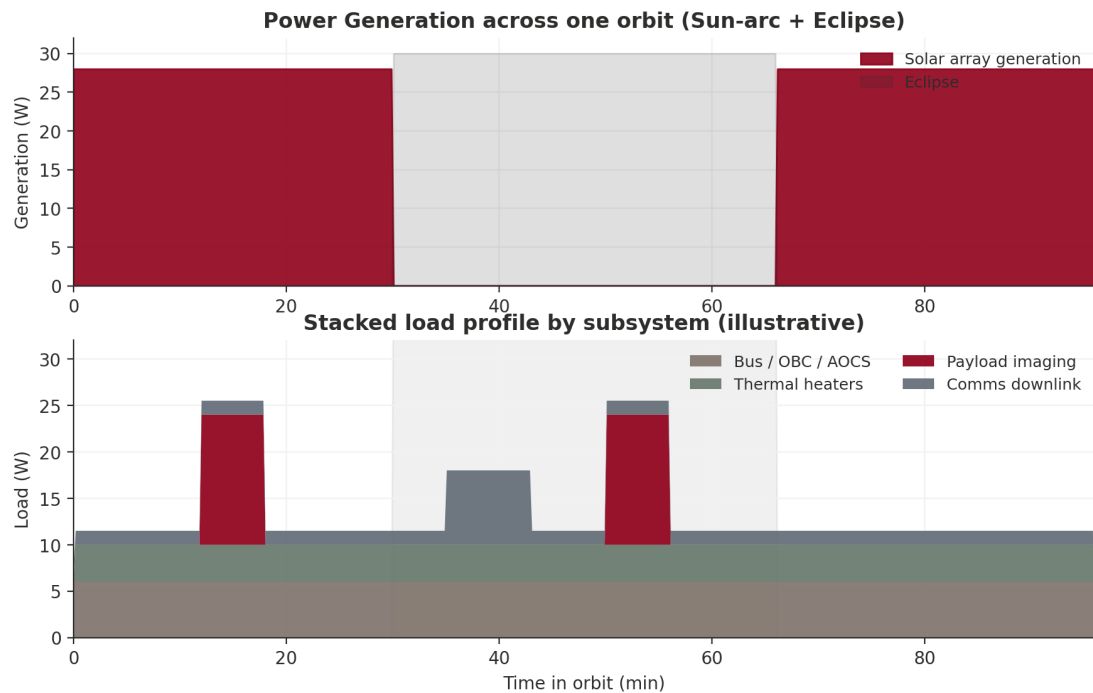
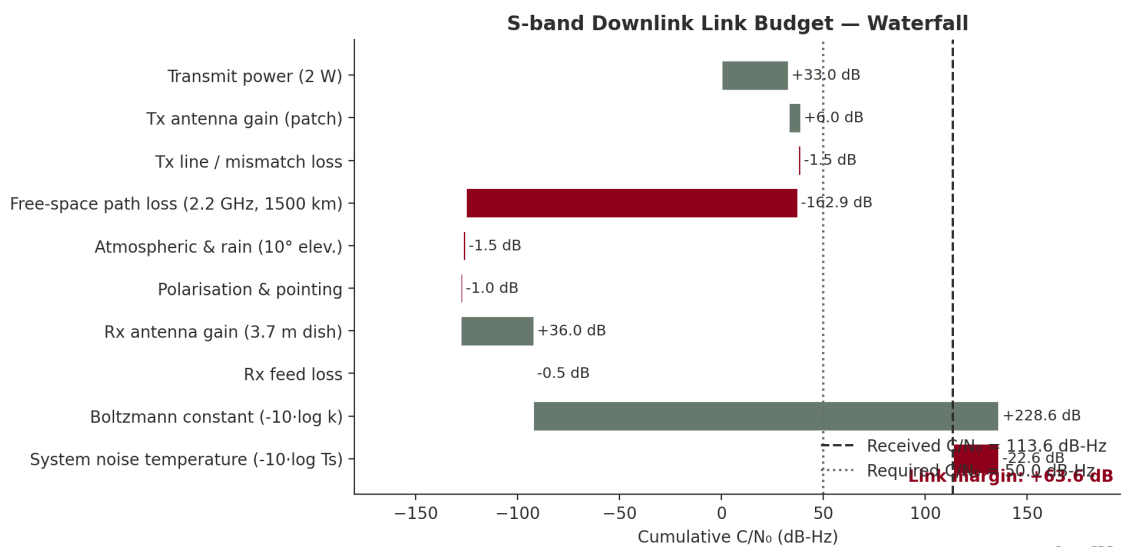


Figure 1.5 — Mass budget and the ECSS margin schedule. Phase A margins start at +44 % above dry mass and tighten phase-by-phase to +5 % at Phase D.



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Figure 1.6 — Power generation and load profile across one orbit. Eclipse is the disciplined consumer of design margin.

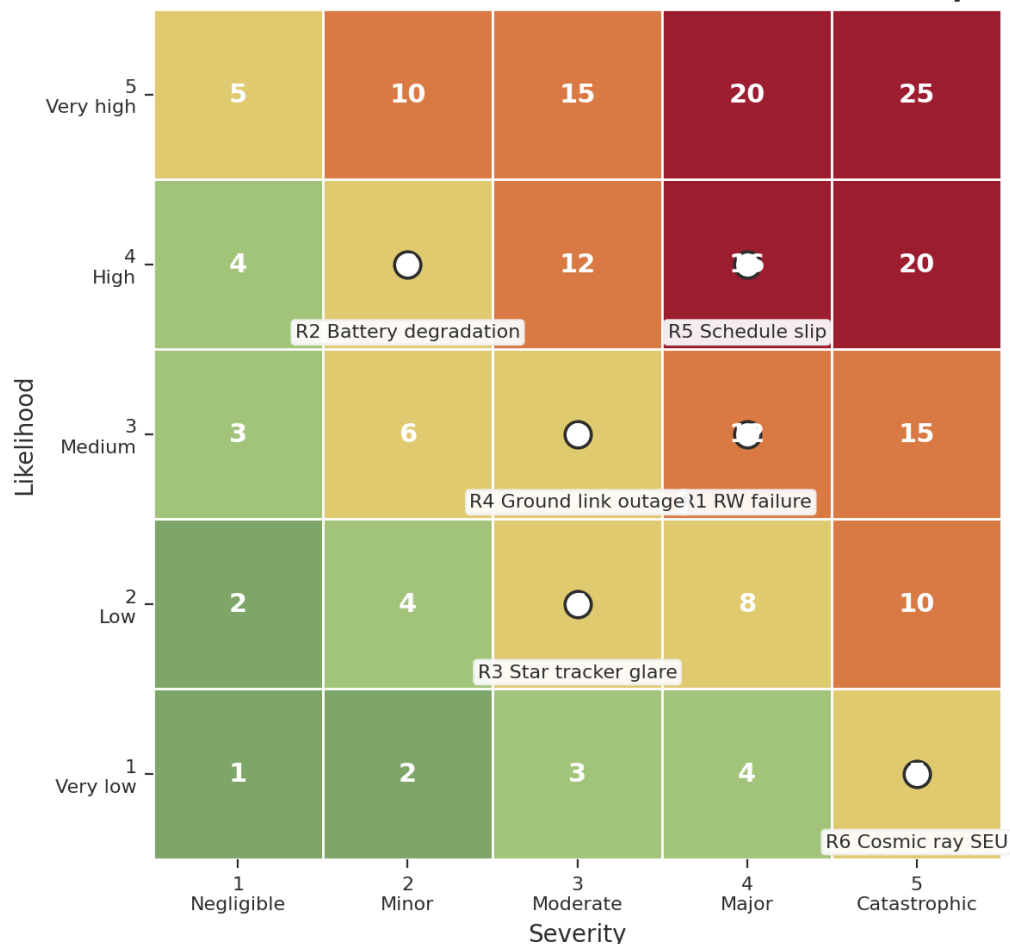


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Figure 1.7 — A link-budget waterfall is the standard way to present a closed link. Required C/N₀ and link margin sit at the bottom-right of every link-budget review.

RISK REGISTER TEMPLATE (WEDNESDAY + THURSDAY)

ID	Title	P (1–5)	I (1–5)	Score	Owner	Mitigation	Status
R1	RW failure mid-life	4	3	12	AOCS	Redundant 4-wheel pyramid; in-flight desat strategy	Open
R2	Battery degradation > 20 %	2	4	8	EPS	Cycle-life model; DoD ≤ 30 % design point	Open
R3	Star-tracker glare during equator pass	3	2	6	AOCS	Two-head ST; baffle design; pointing law	Open
R4	Ground-link outage at LOS	3	3	9	TT&C	Redundant GS (Iqaluit + Troll); SSPA back-off	Open
R5	Programme schedule slip	4	4	16	PM	Critical-path padding; pre-procure long-lead items	Open
R6	Cosmic-ray SEU on OBC	5	1	5	OBDH	Watchdog + EDAC + N-version; daily TM check	Open

Risk Matrix (ECSS-M-ST-80C 5×5) — worked example

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Figure 1.8 — The ECSS-M-ST-80C 5×5 risk matrix. Tier colour coding: 1–2 green, 3–4 light green, 5–9 yellow, 10–15 amber, 16–25 red. R5 (schedule) sits in the amber band; R6 (SEU) is yellow but deserves attention because of the consequence dimension.

EXPECTED READING

Expected reading SMAD4 Ch. 9 (orbit selection) and Ch. 10 (mission analysis) — locally cached on Brightspace. ECSS-E-ST-32C (structures) §4 — <https://ecss.nl/>.

Week 3 — Mission Operations & Simulation

Week 3 puts the cohort into mission-control seats. Two days of operations training, followed by two days running a live simulation on the SpaceCDF / EOSAT-1 simulator at the uOttawa Cyberrange. See the dedicated Week 3 documents for the operations curriculum, procedures, and simulation packs.

Day	Topic	Activities
Mon	Operations Concept Development	Ground-segment architecture. MCS familiarisation (COSMOS / Open-MCT / Yamcs overview). Operations procedures: nominal, contingency, FDIR. Pass planning and scheduling. TC/TM definition from SpaceCDF FSW export.
Tue	Mission Operations Training	Console positions and responsibilities. Voice-loop discipline. Anomaly-response procedures. Dry-run walk-throughs of LEOP, nominal, and contingency scenarios. Practice commanding and telemetry monitoring.
Wed	Mission Simulation Day 1	Full-day simulated LEOP + commissioning. Teams operate their designed mission on console. Deployment, first contact, detumbling, initial checkout. Real-time anomaly injection. Post-sim debrief and lessons-learned.
Thu	Mission Simulation Day 2	Full-day simulated nominal + contingency operations. Science-data acquisition mode. Ground-pass downlink. Injected anomalies: safe-mode entry, power anomaly, comms degradation. Constellation ops (for teams with multi-satellite designs). Post-sim debrief.
Fri	Wrap-Up & Presentations	Final design documentation review. Each team presents complete mission: need, design, ops concept, lessons-learned. Peer review and Q&A. Certificate. Course evaluation. Discussion: next steps, collaboration opportunities, CSA / industry pathways.

Assessment

Component	Weight	Description
Worksheets (daily)	20 %	Completed during sessions, formative feedback
Design Review (Week 2 Fri)	30 %	Team presentation of preliminary design
Simulation Performance (Week 3)	20 %	Console operations, anomaly response
Final Presentation (Week 3 Fri)	30 %	Complete mission design, ops concept, lessons

COMPETENCY SELF-ASSESSMENT GRID

Use this in **week 0** and again at end of **week 3**. Score each row 1 (no exposure) – 5 (could lead a team in this area).

Competency	Wk 0	Wk 3	Δ
Articulate the Canadian space sector and one's place in it			
Identify and apply Canadian regulatory obligations			
Decompose mission objectives to SMART requirements			
Design a mission inside a CDF (any discipline role)			
Maintain a risk register and a planning baseline			
Read & write a procedure file (ECSS-E-ST-70-32C)			
Operate an MCS console under voice-loop discipline			
Lead a structured peer critique			
Synthesise lessons-learned into improvement actions			

Course outcomes

By the end of the programme, every participant will have:

- Worked a complete CubeSat mission design from problem statement to PDR, inside a Concurrent Design Facility.
- Operated each of the six MCS positions for at least one ground-station pass on a live simulator.
- Maintained a risk register and a planning baseline through the full design week.
- Written, run, and critiqued at least one operations procedure.
- Built a peer network across academia, industry, and the Canadian Space Agency.

Reference list

A live, hyperlinked list of the standards, handbooks, and tools referenced in the syllabus.

- **NASA SEH (SP-2016-6105 Rev 2)** — Systems Engineering Handbook. <https://www.nasa.gov/reference/systems-engineering-handbook/>
- **NPR 7123.1D** — NASA Systems Engineering Processes and Requirements. <https://nodis3.gsfc.nasa.gov/>
- **NPR 7120.5F** — NASA Space Flight Program and Project Management Requirements. https://nodis3.gsfc.nasa.gov/displayDir.cfm?Internal_ID=N_PR_7120_005F
- **NASA CubeSat 101** — Basic Concepts and Processes for First-Time CubeSat Developers. https://www.nasa.gov/wp-content/uploads/2017/03/nasa_c-sli_cubesat_101_508.pdf
- **ECSS-E-ST-10C Rev. 1** — System engineering general requirements. <https://ecss.nl/standards/active-standards/ecss-e-st-10c-rev-1-system-engineering-general-requirements/>
- **ECSS-E-ST-10-02C Rev. 1** — Verification. <https://ecss.nl/>
- **ECSS-E-ST-10-24C** — Interface management. <https://ecss.nl/>
- **ECSS-M-ST-10C Rev. 1** — Project planning. <https://ecss.nl/>
- **ECSS-M-ST-80C** — Risk management. <https://ecss.nl/>
- **ECSS-Q-ST-30-02C** — FMEA / FMECA. <https://ecss.nl/>
- **ECSS-E-ST-70C** — Ground systems and operations. <https://ecss.nl/>
- **ECSS-E-ST-70-32C** — Test and operations procedures. <https://ecss.nl/>
- **Cal Poly CDS Rev 14** — CubeSat Design Specification. <https://www.cubesat.org/cds-announcement>
- **CCSDS PUS** — Packet Utilization Standard. <https://public.ccsds.org/Pubs/660x0g3.pdf>
- **ITU-R Radio Regulations** — <https://www.itu.int/pub/R-REG-RR>
- **ISED CPC-2-6-02** — Licensing of space stations. <https://ised-isde.canada.ca/site/spectrum-management-telecommunications/en>
- **Remote Sensing Space Systems Act** — <https://laws-lois.justice.gc.ca/eng/acts/R-5.4/>
- **IADC 25-year debris mitigation guideline** — <https://www.iadc-home.org/>
- **UN COPUOS Registration Convention** — <https://www.unoosa.org/>
- **Wertz, Everett & Puschell** — Space Mission Engineering: The New SMAD (2011).
- **Larson & Wertz** — Space Mission Analysis and Design, 4th ed.
- **Sutton & Biblarz** — Rocket Propulsion Elements, 9th ed.
- **Markley & Crassidis** — Fundamentals of Spacecraft Attitude Determination and Control.
- **Pratt, Bostian & Allnutt** — Satellite Communications, 3rd ed.